Year 11 Mathematics

Number 8

1.3 HCF and LCM

HCF and LCM

STARTER · 5 MIN

Date: _____ **Title:** HCF and LCM

LEARNING OBJECTIVES

- Find the HCF and LCM by listing
- Find the HCF and LCM using prime factors
- Solve problems using HCF and LCM

STARTER *Answer in your book. No calculators.*

1. List the factors of 18
2. List the first five multiples of 6
3. Write 60 as a product of prime factors
4. Write 24 as a product of prime factors

Starter — Answers

STARTER

Tick or correct in green pen

1. **1, 2, 3, 6, 9, 18**

2. **6, 12, 18, 24, 30**

3. $60 = \mathbf{2^2 \times 3 \times 5}$

4. $24 = \mathbf{2^3 \times 3}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the HCF and LCM by listing	☐
SILVER	I can find the HCF and LCM using prime factors	☐
GOLD	I can solve problems using HCF and LCM	☐

Tick each box in the plenary when you can do it on your own.

Introduction — HCF and LCM

INTRODUCTION · 3 MIN

HCF: the biggest number that divides both. LCM: the smallest number both divide into.

HCF

Highest Common Factor.

HCF of 12 and 18 = 6

LCM

Lowest Common Multiple.

LCM of 4 and 6 = 12

HCF from primes

Multiply the primes they share (lowest powers).

$2^2 \times 3$ and 2×3^2: HCF = 2×3

LCM from primes

Multiply all primes (highest powers).

LCM = $2^2 \times 3^2 = 36$

Examples 1 and 2 — Listing Method

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Find the HCF of 16 and 24.

Example 2

Find the LCM of 6 and 8.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

16: 1, 2, 4, 8, 16

24: 1, 2, 3, 4, 6, 8, 12, 24

HCF = 8

Tip: find the biggest factor in both lists.

Example 2

6: 6, 12, 18, 24

8: 8, 16, 24

LCM = 24

Tip: find the first multiple in both lists.

Task 1

PRACTICE · 9 MIN

BRONZE *Answer in your book*

1. HCF of 12 and 30
2. HCF of 20 and 35
3. LCM of 4 and 10
4. LCM of 9 and 12
5. HCF of 36 and 48

★ LCM of 5, 6 and 10

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. HCF = **6**
2. HCF = **5**
3. LCM = **20**
4. LCM = **36**
5. HCF = **12**

★ LCM = **30**

Examples 3 and 4 — Using Prime Factors

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint
HCF: shared primes, lowest powers
LCM: all primes, highest powers

Example 3

$60 = 2^2 \times 3 \times 5, \quad 72 = 2^3 \times 3^2$

Find the HCF of 60 and 72.

Example 4

Find the LCM of 60 and 72.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Shared primes: 2^2 and 3

HCF = $2^2 \times 3$

= 12

Tip: use the lowest power of each shared prime.

Example 4

Highest powers: 2^3, 3^2, 5

LCM = $8 \times 9 \times 5$

= 360

Tip: use every prime, highest power.

Task 2

PRACTICE · 9 MIN

SILVER

Use prime factors to find the HCF and LCM

Hint
HCF: shared primes, lowest powers
LCM: all primes, highest powers

1. 18 and 30
2. 24 and 36
3. 40 and 100
4. 45 and 75
5. 84 and 126

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. HCF = 6, LCM = **90**
2. HCF = 12, LCM = **72**
3. HCF = 20, LCM = **200**
4. HCF = 15, LCM = **225**
5. HCF = 42, LCM = **252**

Examples 5 and 6 — HCF and LCM Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint
HCF: shared primes, lowest powers
LCM: all primes, highest powers

Example 5

(3 marks)

Buses leave every 12 minutes and trains every 20 minutes.

Both leave at 9:00. When do they next leave together?

Example 6

(3 marks)

Ribbons are 48 cm and 60 cm long.

They are cut into equal pieces, as long as possible.

How long is each piece?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

LCM of 12 and 20 = 60

60 minutes later

10:00

Tip: 'together again' means LCM.

Example 6

HCF of 48 and 60

= 12

Each piece **= 12 cm**

Tip: 'as large as possible' means HCF.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint HCF: shared primes, lowest powers
LCM: all primes, highest powers

1. Lights flash every 8 s and every 14 s. They flash together. When do they next flash together? (3 marks)

2. 36 boys and 54 girls are split into equal groups with no mixing. What is the largest group size? (3 marks)

3. Cups come in packs of 6 and plates in packs of 15. What is the smallest number of each so they match? (3 marks)

4. $A = 2^3 \times 3 \times 5^2$, $B = 2 \times 3^2 \times 5$. Find the HCF and LCM of A and B. (3 marks)

5. The HCF of two numbers is 6 and the LCM is 36. One number is 12. Find the other. (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. LCM = 56, **56 seconds**

2. HCF = 18, **18 students**

3. LCM = 30, **30 cups and 30 plates**

4. $\text{HCF} = 2 \times 3 \times 5 = 30, \quad \text{LCM} = 2^3 \times 3^2 \times 5^2 = \mathbf{1800}$

5. $6 \times 36 \div 12 = \mathbf{18}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find the HCF of 15 and 25

SILVER Find the LCM of 8 and 12

GOLD Find the HCF of $2^2 \times 3 \times 7$ and $2 \times 3^2 \times 7$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE HCF = **5**

SILVER LCM = **24**

GOLD HCF = $2 \times 3 \times 7$ = **42**

Exam Practice — HCF and LCM

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1 (2 marks)

Find the highest common factor (HCF) of 28 and 42.

.....

Question 2 (2 marks)

Find the lowest common multiple (LCM) of 8 and 10.

.....

Exam Practice — Using Prime Factors

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint
HCF: shared primes, lowest powers
LCM: all primes, highest powers

Question 3

(2 marks)

$90 = 2 \times 3^2 \times 5$

$84 = 2^2 \times 3 \times 7$

Find the HCF of 90 and 84.

.....

Question 4

(2 marks)

$90 = 2 \times 3^2 \times 5$

$84 = 2^2 \times 3 \times 7$

Find the LCM of 90 and 84.

.....

Exam Practice — HCF and LCM Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint
HCF: shared primes, lowest powers
LCM: all primes, highest powers

Question 5

(3 marks)

Two lighthouses flash together at midnight.

One flashes every 18 seconds.

The other flashes every 24 seconds.

How many seconds after midnight do they next flash together?

..... seconds

Question 6

(3 marks)

$A = 2^3 \times 3^2 \times 5$

$B = 2 \times 3^3 \times 7$

(a) Find the HCF of A and B. (1)

.....

(b) Find the LCM of A and B. (2)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

28: 1, 2, 4, 7, 14, 28 — M1

42: 1, 2, 3, 6, 7, 14, 21, 42

HCF = 14 — A1

Tip: the biggest factor in both lists.

Question 2

8: 8, 16, 24, 32, 40 — M1

10: 10, 20, 30, 40

LCM = 40 — A1

Tip: the first multiple in both lists.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

shared primes: 2 and 3 — M1

HCF = 2 × 3 = 6 — A1

Tip: lowest power of each shared prime.

Question 4

$2^2 \times 3^2 \times 5 \times 7$ — M1

LCM = 1260 — A1

Tip: every prime, highest power.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

LCM of 18 and 24 — M1

$18 = 2 \times 3^2$, $24 = 2^3 \times 3$ — M1

LCM = 72 seconds — A1

Tip: 'together again' means LCM.

Question 6

(a) $2 \times 3^2 = 18$ — B1

(b) $2^3 \times 3^3 \times 5 \times 7$ — M1

= 7560 — A1

Tip: leave the answer in index form if allowed.